Smart Item Bank: Database Design

PostgreSQL schema (with the pgvector extension) for the React + FastAPI + LangChain/LangGraph build.

Design Principles

- **One question bank, many sources.** AI-generated and teacher-written questions live in the same `questions` table, separated by a `source` column, so quizzes and exports treat them identically.
- **Everything hangs off course and topic.** Materials, questions, assessments and student searches all carry a `course_id` (and `topic_id` where relevant), so filtering and analytics stay simple.
- **Review before reuse.** A question is only usable in quizzes once its `status` is `approved`, which is how the teacher review step is enforced.
- **UUID primary keys** for all tables, with `created_at` and `updated_at` timestamps, and soft deletes (`deleted_at`) on content a teacher might delete by mistake.
- **Search logging must be cheap.** `search_logs` is append-only with minimal indexes so recording a search never slows the student's response.
- **LangGraph checkpoints** are stored by the LangGraph Postgres checkpointer in its own tables. The app only keeps a `generation_runs` row that points to the LangGraph `thread_id`.

Entity Overview

Area	Tables
Users and access	`users`, `course_enrollments`
Course structure	`courses`, `topics`
Materials	`materials`, `material_chunks`
AI pipeline	`generation_runs`
Question bank	`questions`, `question_options`, `tags`, `question_tags`
Assessments	`assessments`, `assessment_questions`, `attempts`, `attempt_answers`

Area	Tables
Exports	`exports`
Search and insights	`search_logs`, `search_clusters`, `search_cluster_members`, `suggested_actions`

Enumerated Types

Enum	Values
`user_role`	`admin`, `teacher`, `student`
`question_type`	`mcq`, `short_answer` (more later: `true_false`, `long_answer`)
`question_source`	`ai`, `manual`
`question_status`	`draft`, `approved`, `rejected`, `archived`
`difficulty`	`easy`, `medium`, `hard`
`material_status`	`uploaded`, `extracting`, `ready`, `failed`
`run_status`	`queued`, `running`, `awaiting_review`, `completed`, `failed`
`assessment_status`	`draft`, `published`, `closed`
`attempt_status`	`in_progress`, `submitted`, `graded`
`action_status`	`open`, `done`, `dismissed`

Users and Access

users

Column	Type	Notes
id	uuid, PK	
email	citext, unique	Login identifier
password_hash	text	Never store plain passwords
full_name	text	

Column	Type	Notes
role	user_role	Drives permissions
is_active	boolean	Default true
created_at, updated_at	timestamptz	

course_enrollments

Links students to the courses they can see and take quizzes in.

Column	Type	Notes
id	uuid, PK	
course_id	uuid, FK courses	
student_id	uuid, FK users	Must have role `student`
enrolled_at	timestamptz	

Unique on (`course_id` , `student_id`).

Course Structure

courses

Column	Type	Notes
id	uuid, PK	
code	text	For example `CS301`
title	text	For example "Database Systems"
description	text	Optional
teacher_id	uuid, FK users	Owning teacher
created_at, updated_at, deleted_at	timestamptz	

If several teachers share a course later, add a `course_teachers` join table and drop `teacher_id` .

topics

Column	Type	Notes
id	uuid, PK	
course_id	uuid, FK courses	
parent_topic_id	uuid, FK topics, nullable	Allows sub-topics later
title	text	For example "SQL Joins"
position	int	Display order
created_at, updated_at, deleted_at	timestamptz	

Unique on (`course_id` , `title` , `parent_topic_id`).

Materials

materials

One row per uploaded PDF, PPTX or DOCX.

Column	Type	Notes
id	uuid, PK	
course_id	uuid, FK courses	
topic_id	uuid, FK topics	
uploaded_by	uuid, FK users	
file_name	text	Original name
file_type	text	`pdf` , `pptx` , `docx`
storage_path	text	Local path or object-storage key
file_size_bytes	bigint	
content_hash	text	Detects duplicate uploads
status	material_status	Extraction progress
error_message	text, nullable	Filled when extraction fails

Column	Type	Notes
extracted_text	text, nullable	Full cleaned text (optional, can be rebuilt from chunks)
created_at, updated_at, deleted_at	timestamptz	

material_chunks

Text split for AI generation and for semantic search. This is where the pgvector embeddings live.

Column	Type	Notes
id	uuid, PK	
material_id	uuid, FK materials	Cascade delete
course_id	uuid, FK courses	Denormalised for fast filtering
topic_id	uuid, FK topics	Denormalised
chunk_index	int	Order within the material
page_or_slide	int, nullable	Source location
content	text	
token_count	int	
embedding	vector(N)	N depends on the embedding model chosen

Indexes: HNSW index on `embedding` (cosine), plus btree on (`course_id`, `topic_id`).

AI Pipeline

generation_runs

One row each time a teacher asks the AI to generate questions. It ties the LangGraph run to the questions it produced and records cost.

Column	Type	Notes
id	uuid, PK	

Column	Type	Notes
course_id	uuid, FK courses	
topic_id	uuid, FK topics	
requested_by	uuid, FK users	
material_ids	uuid[]	Materials used as the source
settings	jsonb	Counts, types, difficulty mix, instructions
langgraph_thread_id	text	Links to the checkpointer so a paused run can resume
status	run_status	`awaiting_review` while the teacher reviews
model_name	text	LLM used
prompt_version	text	Which prompt template was used
input_tokens, output_tokens	int	For cost tracking
estimated_cost	numeric(10,4)	
error_message	text, nullable	
created_at, completed_at	timestamptz	

Question Bank

questions

Column	Type	Notes
id	uuid, PK	
course_id	uuid, FK courses	
topic_id	uuid, FK topics	
created_by	uuid, FK users	The teacher who created or approved it
source	question_source	`ai` or `manual`

Column	Type	Notes
generation_run_id	uuid, FK generation_runs, nullable	Only for AI questions
source_chunk_id	uuid, FK material_chunks, nullable	Which text the AI used, useful for verification
type	question_type	
stem	text	The question text
correct_answer_text	text, nullable	For short-answer questions, the reference answer
explanation	text, nullable	Shown to students after grading
difficulty	difficulty	Nullable at first
marks	numeric(5,2)	Default 1
status	question_status	Only `approved` questions can go into quizzes
times_used	int	Updated when added to an assessment
search_vector	tsvector	Full-text search on stem and explanation
created_at, updated_at, deleted_at	timestamptz	

Indexes: (`course_id` , `topic_id` , `status`), GIN on `search_vector` .

question_options

Answer choices for MCQs.

Column	Type	Notes
id	uuid, PK	
question_id	uuid, FK questions	Cascade delete
label	char(1)	A, B, C, D
text	text	

Column	Type	Notes
is_correct	boolean	
position	int	

Rule enforced in the API: an MCQ has exactly one correct option (a partial unique index on (`question_id`) where `is_correct` is true).

tags and question_tags

Optional labels such as "exam-ready" or "chapter 3".

Table	Columns
tags	id (uuid, PK), course_id (FK), name (text), unique on (`course_id`, `name`)
question_tags	question_id (FK), tag_id (FK), primary key on both

Assessments

assessments

Column	Type	Notes
id	uuid, PK	
course_id	uuid, FK courses	
created_by	uuid, FK users	
title	text	
instructions	text, nullable	
kind	text	`quiz` or `exam`
duration_minutes	int, nullable	
available_from, available_until	timestamptz	Availability window
max_attempts	int	Default 1
shuffle_questions	boolean	
status	assessment_status	

Column	Type	Notes
created_at, updated_at, deleted_at	timestamptz	

assessment_questions

The questions chosen for an assessment, in order.

Column	Type	Notes
id	uuid, PK	
assessment_id	uuid, FK assessments	
question_id	uuid, FK questions	
position	int	
marks	numeric(5,2)	Overrides the question's default if set

Unique on (`assessment_id` , `question_id`).

attempts

Column	Type	Notes
id	uuid, PK	
assessment_id	uuid, FK assessments	
student_id	uuid, FK users	
attempt_number	int	
status	attempt_status	
started_at, submitted_at	timestamptz	
score	numeric(6,2), nullable	
max_score	numeric(6,2)	

Unique on (`assessment_id` , `student_id` , `attempt_number`).

attempt_answers

Column	Type	Notes
id	uuid, PK	
attempt_id	uuid, FK attempts	
question_id	uuid, FK questions	
selected_option_id	uuid, FK question_options, nullable	For MCQs
answer_text	text, nullable	For short answers
is_correct	boolean, nullable	Set by auto-grading
awarded_marks	numeric(5,2), nullable	
ai_feedback	text, nullable	From the optional AI grading workflow
graded_by	uuid, FK users, nullable	Set when a teacher overrides the AI

Unique on (`attempt_id`, `question_id`), which also supports autosave updates.

Exports

exports

A small log of Word exports so teachers can re-download and you can see usage.

Column	Type	Notes
id	uuid, PK	
requested_by	uuid, FK users	
course_id	uuid, FK courses	
question_ids	uuid[]	Questions included
include_answer_key	boolean	
storage_path	text, nullable	Generated DOCX, if kept
created_at	timestamptz	

Student Search and Teacher Insights

search_logs

Append-only record of every student search.

Column	Type	Notes
id	bigserial, PK	
student_id	uuid, FK users	
course_id	uuid, FK courses, nullable	
topic_id	uuid, FK topics, nullable	
query_text	text	What the student typed
normalized_query	text	Lowercased and trimmed, used for quick grouping
query_embedding	vector(N), nullable	Filled by a background job for clustering
result_count	int	
searched_at	timestamptz	

Indexes: (`course_id`, `searched_at`), (`course_id`, `normalized_query`). Consider partitioning by month if volume grows.

search_clusters

Groups of similar searches produced by the LangGraph analytics workflow (for example "LEFT JOIN" and "left outer join").

Column	Type	Notes
id	uuid, PK	
course_id	uuid, FK courses	
topic_id	uuid, FK topics, nullable	
label	text	Short name chosen by the AI, for example "LEFT JOIN"

Column	Type	Notes
search_count	int	
unique_students	int	
first_seen_at, last_seen_at	timestamptz	
computed_at	timestamptz	When the analytics workflow last updated it

search_cluster_members

Column	Type	Notes
cluster_id	uuid, FK search_clusters	
search_log_id	bigint, FK search_logs	Primary key on both

suggested_actions

The "students keep searching X, create more questions" suggestions shown to the teacher.

Column	Type	Notes
id	uuid, PK	
course_id	uuid, FK courses	
cluster_id	uuid, FK search_clusters	
action_type	text	`generate_questions`, `add_material`, `review_quiz`
message	text	Teacher-facing text
status	action_status	
created_at, resolved_at	timestamptz	
generation_run_id	uuid, FK generation_runs, nullable	Set if the teacher clicked through to generate questions

Key Relationships

- A **course** has many **topics**, **materials**, **questions**, **assessments** and **enrollments**.
- A **topic** has many **materials** and **questions**.
- A **material** has many **material_chunks**, and AI questions point back to the chunk they came from.
- A **generation_run** produces many **questions**, which stay `draft` until the teacher approves or rejects them.
- An **assessment** links to **questions** through **assessment_questions**, and each student **attempt** has many **attempt_answers**.
- **search_logs** feed **search_clusters**, which produce **suggested_actions**, which can launch a new **generation_run**.

Suggested Build Order

This matches the project milestones, so each migration lands with the feature that needs it.

Milestone	Tables to create
1. Foundation	`users`, `courses`, `topics`, `course_enrollments`, `materials`
2. AI generation and item bank	`material_chunks`, `generation_runs`, `questions`, `question_options`, `tags`, `question_tags`
3. Quizzes and attempts	`assessments`, `assessment_questions`, `attempts`, `attempt_answers`, `exports`
4. Search and insights	`search_logs`, `search_clusters`, `search_cluster_members`, `suggested_actions` (pgvector indexes on chunks and searches come with this milestone)

Open Decisions

- **Embedding model and vector size:** this fixes the dimension in `vector(N)`, and changing it later means re-embedding everything.
- **File storage:** local disk for the prototype versus S3-style storage for production affects `storage_path` handling only.

- **Multiple teachers per course:** add `course_teachers` now if this is likely, since it is harder to change later.
- **Question versioning:** if edited questions must not change past quizzes, add a `question_versions` table and point `assessment_questions` at a version instead of the question.
- **Student identity in search logs:** decide whether teachers see student names or only anonymised counts, which may matter for privacy policy.